SUPPLEMENTAL VIDEO REVIEW

Check Your Knowledge Questions and Explanations

Pharmacology of Diuretic Agents, Part 2

- **Thiazide Diuretics**
- **Potassium-Sparing Diuretics**
  - ENaC Inhibitors
  - Mineralocorticoid Receptor Antagonists
- **Osmotic Diuretics**

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Diuretics Pharmacology is one lecture in two parts. The learning objectives are the same for both.

After completing the preparation materials, students should be able to:

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1. **Sites of action:** Map each diuretic class to its site of action in the nephron and predict the downstream effects.
2. **Mechanisms:** Explain how inhibition of specific renal transporters alters sodium handling, electrochemical gradients and downstream nephron function to produce diuresis.
3. **Effects:**
   - Correlate each diuretics class with their urinary electrolyte patterns.
   - Predict acid-base disturbances based on mechanisms.
   - Explain diuretic effects on tubuloglomerular feedback, renal hemodynamics, GFR.
4. **Compensatory responses:** Relate diuretic actions to compensatory physiological responses (e.g., SNS, RAAS activation, distal nephron reabsorption) and explain their role in diuretic resistance.
5. **Therapeutic uses:** Select the appropriate diuretic based on mechanism, patient factors, and clinical scenario.
6. **Clinical considerations:**
   - Explain how the pharmacokinetic properties influence onset, efficacy, and clinical use of the diuretic agents.
   - Predict the adverse effects and contraindications based on diuretic mechanism and patient factors.
   - Predict the impact of NSAIDs on diuretic response by describing the role of the renal prostaglandins on the actions and effects of the diuretic agents.
   - Predict drug interactions based on the mechanism of action, pharmacokinetic properties, or pharmacogenetic variabilities of the individual drugs or drug classes

Preparation Materials

Required

- Dr. Goldstein's Practice Questions

Practice questions are valuable for learning because they force the brain to retrieve information, which strengthens neural pathways, interrupts the forgetting curve, and makes knowledge significantly easier to recall.

Optional materials:

Use the resources that work best for you. You don't need to study all of them.

- Lecture Videos | Dr. Goldstein's Word handout
- ScholarRx Bricks | Guided Reading Questions
- Textbook resources with links are listed on the next slide.
- Review books and ScholarRx have more practice questions.
- Write your own questions – what you would like to learn more about.

Links to resources

Scholar Rx Bricks: Renal https://exchange.scholarrx.com/collection/renal

Renal> Treatment of Renal Disorders > Diuretics

Diuretics Foundations and Frameworks https://exchange.scholarrx.com/brick/diuretics-foundations-and-frameworks

Carbonic Anhydrase Inhibitors https://exchange.scholarrx.com/brick/carbonic-anhydrase-inhibitors

Loop Diuretics https://exchange.scholarrx.com/brick/loop-diuretics

Thiazide Diuretics https://exchange.scholarrx.com/brick/thiazide-diuretics

Potassium-Sparing Diuretics https://exchange.scholarrx.com/brick/potassium-sparing-diuretics

Osmotic Diuretics https://exchange.scholarrx.com/brick/osmotic-diuretics

Textbooks/Exam review books you might find helpful:

Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 15: Diuretic Agents
https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3058&sectionid=255304977

Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 15: Diuretics and Other Drugs That Act on the Kidney
https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285591587#1207298272

LWW Health Library Premium Basic Sciences; Lippincott's Illustrated Reviews: Pharmacology 8e, 2023; Chapter 9: Diuretics

https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253326102&bookid=3222

Key points: Thiazide diuretics

- Thiazide-type and thiazide-like diuretics (broadly categorized as thiazide diuretics) act on the sodium chloride cotransporter (NCC) in the distal convoluted tubule where ~5-10% of sodium is reabsorbed. Blocking NCC leads to increased sodium, chloride, and water excretion.
- Thiazide diuretics cause mild diuresis, which reduces plasma volume and decreases cardiac output, resulting in lower blood pressure.
- With continued therapy, the plasma volume returns to pretreatment levels (a desirable effect), with long-term antihypertensive effects.
- Thiazide diuretics are recommended first-line agents for the management of primary hypertension and are useful in the treatment of Liddle syndrome and vasopressin resistance (nephrogenic diabetes insipidus). (vasopressin = antidiuretic hormone)
- Thiazide diuretics cause potassium-wasting effect, mild hypokalemic metabolic alkalosis, and hyponatremia and are associated with increased serum glucose, lipids, and uric acid.
- Thiazides increase urinary magnesium loss by impairing the apical magnesium channel (TRPM6) in the DCT.
- Students should be aware of specific adverse effects and drug interactions.

Key points: Potassium-sparing diuretics

- Epithelial sodium channel (ENaC) inhibitors and mineralocorticoid receptor antagonists (MRAs) are potassium-sparing diuretics that inhibit sodium reabsorption in the principal cells located in the late DCT, connecting tubule, and collecting ducts. Sodium reabsorption in the principal cells is tightly regulated by aldosterone (aldosterone-sensitive sites).
- They are considered weak diuretics because they block reabsorption of a small percentage of filtered sodium (~2-5%).
- They are known as the potassium-sparing diuretics because they prevent loss of potassium in the urine. There is a risk of hyperkalemia especially when administered with other potassium-retaining drugs.
- Potassium-sparing diuretics are typically given with another diuretic for additive effect and to offset potassium loss.
- ENaC inhibitors, amiloride and triamterene, reduce the passive apical entry of sodium ions, which reduces the lumen-negative transepithelial potential.
- The MRAs, spironolactone, eplerenone, and finerenone, block the actions of aldosterone, an adrenal hormone that promotes sodium reabsorption and potassium excretion.
- Students should be aware of specific adverse effects and drug interactions.

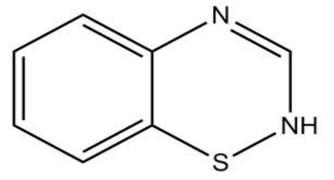
Key points: Osmotic diuretics

- **Kidney:** Osmotic diuretics are freely filtered at the glomerulus, not reabsorbed, and are pharmacologically inert (no receptor targets). They stay in the tubular lumen, increasing the osmolarity of the luminal fluid. The greatest effect is in the water permeable segments – the proximal tubule (major effect) and the descending limb of the loop of Henle.
- The increased luminal flow reduces passive Na+ reabsorption and substantially increases urine output, producing rapid diuresis.
- **Plasma:** Osmotic diuretics, such as mannitol, stay in extracellular fluid. Plasma osmolality rises, causing water to shift out of cells into the plasma.
- **Therapeutic uses:** By increasing the osmotic pressure of plasma, osmotic diuretics extract water from the eye and brain. Osmotic diuretics are used to control intraocular pressure during acute attacks of glaucoma and for short-term reductions in intraocular pressure both preoperatively and postoperatively in patients who require ocular surgery. Also, mannitol is used to reduce cerebral edema and brain mass before and after neurosurgery and to control intracranial pressure in patients with traumatic brain injury.
- **Adverse effects:** Osmotic diuretics can cause pulmonary edema or worsen heart failure.

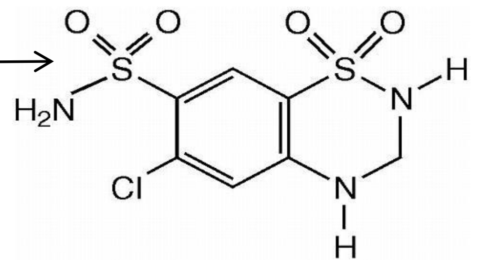
THIAZIDE DIURETICS
Na^+-Cl^- Cotransporter (NCC) Inhibitors

Thiazide-type (benzothiadiazine derivatives)	Thiazide-like (lack benzothiadiazine structure)
✳Hydrochlorothiazide	✳Chlorthalidone
Chlorothiazide	Indapamide
Methyclothiazide Bendroflumethiazide — Rarely used	Metolazone

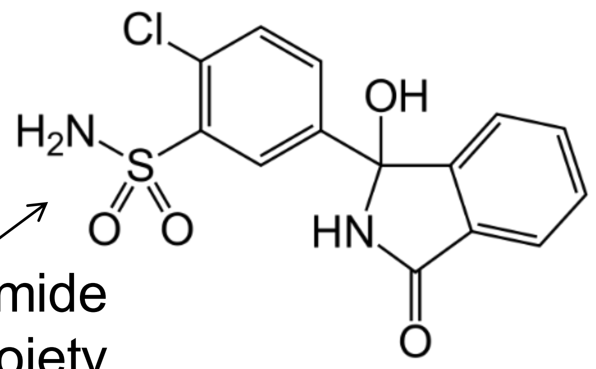
Benzothiadiazine



sulfonamide moiety



Hydrochlorothiazide



sulfonamide moiety

Chlorthalidone

Boards-relevant drug-specific features of indapamide and metolazone:
Indapamide has a unique vasodilatory action and stronger BP reduction at low doses.
Metolazone is effective when GFR is reduced, has proximal tubule activity, powerful synergy with loop diuretics, more intense electrolyte disturbances, but is not suitable for routine hypertension management.

Pharmacokinetics of selected thiazide diuretics

PK Takeaway: Well absorbed, bioavailability generally good, moderate protein binding, most are not metabolized, enter nephron via filtration and OAT secretion, excreted as intact drug, half-lives vary

Drug	Oral Bioavailability	Onset of diuresis Duration of action	Route of elimination	t½ elimination (approximate)
HCTZ hydrochlorothiazide	70%	2 hours 6-12 hours	Urine	6-15 hours
Chlorthalidone	65%	2-6 hours 24-72 hour	Urine 50-65% Hepatic metabolism and fecal excretion	40-60 hours

Chlorthalidone Takeaway: Very long t½, long duration of action provides smoother 24-hour blood pressure control. Higher risk of hypokalemia than with HCTZ

Plasma protein binding varies considerably among thiazide diuretics. Hydrochlorothiazide, chlorothiazide, and chlorthalidone are approximately 70-75% plasma protein bound at therapeutic doses.

t½ adults, normal renal function

Chlorothiazide is the only intravenous thiazide diuretic.

Indapamide is extensively metabolized in the liver and has a long half-life.

Na^+-Cl^- cotransporter (NCC) inhibition → ↑ Na^+,Cl^- excretion and moderate diuresis

NCC inhibition in the distal convoluted tubule (DCT)

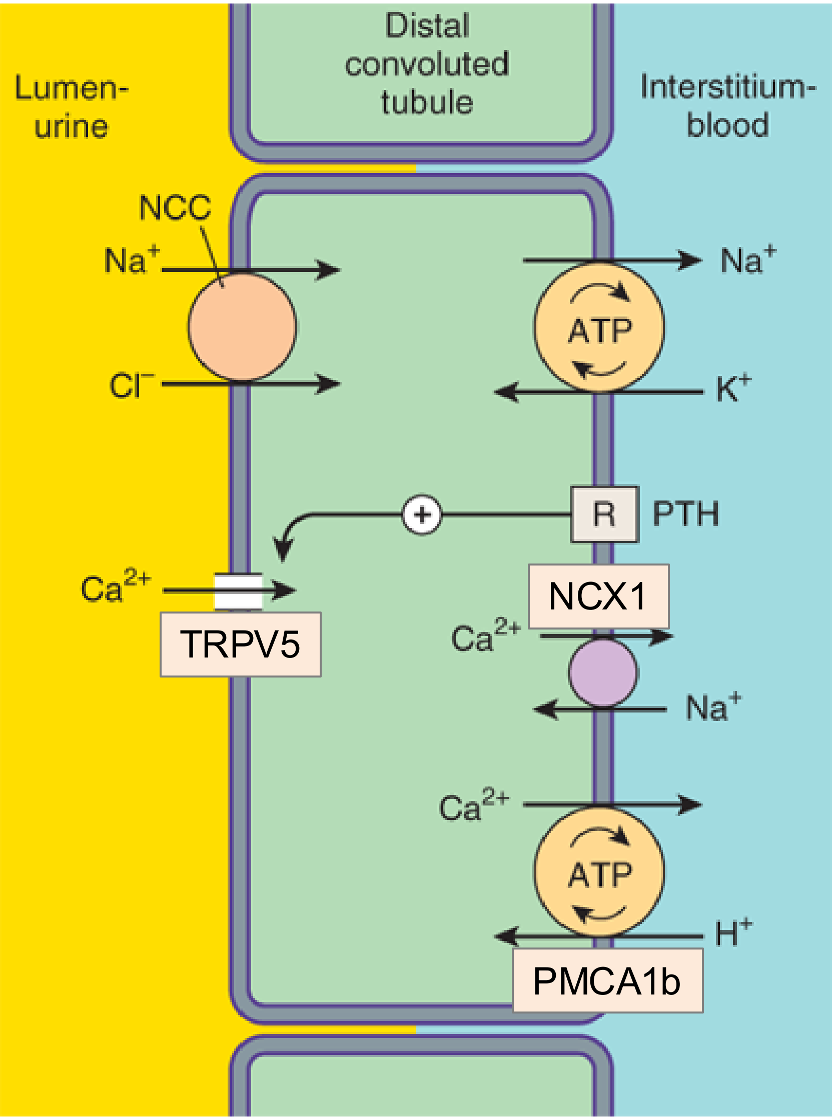
↓

↑ Na^+,Cl^- excretion

↓

moderate diuresis

- ~5-10% of filtered Na^+ is reabsorbed in the early DCT.
- Moderate diuretic efficacy



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

Legend:
Ion transport pathways across the luminal and basolateral membranes of the distal convoluted tubule cell. As in all tubular cells, Na^+/K^+-ATPase is present in the basolateral membrane. NCC is the primary sodium and chloride transporter in the luminal membrane. R, parathyroid hormone (PTH) receptor.

Slide adapted by LG

Thiazide diuretics increase local calcium export in the DCT due to NCC blockade.

Mechanism:

Altered Na^+ handling and increased activity of Ca^{2+} transport pathways:

- Blocking luminal NCC in the DCT lowers intracellular Na^+ concentration→ enhances basolateral NCX1 activity (3Na^+ in, 1 Ca^{2+} out).
- Low intracellular Ca^{2+} creates stronger gradient for Ca^{2+} entry via TRPV5 → ↑Ca^{2+} into the cell → exported out via basolateral NCX1
- PMCA1b (ATPase) fine-tunes Ca^{2+} export.

Physiology:

Active Ca^{2+} reabsorption in the DCT is regulated by parathyroid hormone, PTH, and vitamin D.

TRPV5 is a highly calcium-selective voltage-dependent ion channel expressed on the apical membrane. It regulates transcellular calcium reabsorption.
PMCA1b: Plasma membrane calcium ATPase, isoform 1b. It maintains very low intracellular calcium levels.

Citation: Chapter 15 Diuretic Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021 Figure 15-4. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=250596112&bookid=2988 Accessed: July 22, 2021

Summary of Changes in Urinary Electrolyte Patterns and Body pH in Response to CA Inhibitors, Loop Diuretics, and Thiazide Diuretics

	Urinary Electrolytes			
Diuretic Class	Na^+	HCO_3^-	K^+	Body pH
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop diuretics	++++	0	+++	↑
Thiazides	++	+	++	↑
Loop diuretics plus thiazides	+++++	+	++++	↑

+, increase; –, decrease; 0, no change; ↓, acidosis; ↑, alkalosis

RECAP: Effects of Thiazide Diuretics
↑ Na^+ excretion ~5-10% of filtered sodium; water follows → moderate natriuresis / diuresis
K+ loss/hypokalemia, increased H+ secretion, ECF volume contraction / RAAS activation → hypokalemic metabolic alkalosis. Hypochloremia and bicarbonate retention sustain metabolic alkalosis.
Thiazide diuretics increase Ca^{2+} reabsorption and moderate Mg^{2+} loss.
Thiazide diuretics impair the kidney's ability to excrete dilute urine, which impairs excretion of free water. Thiazides do NOT impair the kidney's ability to concentrate urine, which can lead to hyponatremia.
Thiazides can produce hyperglycemia and insulin resistance, mild dyslipidemia, and increased lithium levels.

Thiazide Diuretics: Clinical Applications

<table>
<tr>
<td>Hypertension
Thiazide diuretics are first-line:
Effective, consistent BP lowering effect with stronger outcome data than several other classes of antihypertensive agents.</td>
<td>• Acutely lower BP by reducing ECF and cardiac output.
• Sustained effect is due to chronic sodium depletion that decreases vascular smooth muscle responsiveness to vasoconstrictors → persistent reduction in peripheral vascular resistance.
Hydrochlorothiazide is most widely used – in many fixed-combo products.
Chlorthalidone has a much longer t½.
Comparative efficacy: Large-scale observational studies and randomized clinical trials have shown that both drugs reduce blood pressure and decrease the risk of heart attacks, stroke, and heart failure at similar rates (Ishani, Cushman, et.al.; 2022).</td>
</tr>
<tr>
<td>Congestive heart failure</td>
<td rowspan="5">Edematous states
Sequential nephron blockade:
Thiazide combined with a loop diuretic for synergy.</td>
</tr>
<tr><td>Liver cirrhosis (ascites)</td></tr>
<tr><td>Nephrotic syndrome</td></tr>
<tr><td>Chronic renal failure</td></tr>
<tr><td>Acute glomerulonephritis</td></tr>
</table>

Thiazide Diuretics: Clinical Applications

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Vasopressin resistance (AVP-R), including lithium-induced AVP-R (nephrogenic diabetes insipidus) TZD + low sodium, low protein diet + NSAID	In AVP-R, the collecting duct cannot respond to ADH. NCC blockade by thiazide diuretics shifts Na+ reabsorption upstream. Mechanism: Thiazide-induced mild volume contraction → RAAS activation and ↓GFR → Na^+ and water reabsorption in proximal tubule → substantial drop in filtrate reaching ADH-sensitive sites in collecting ducts → overall urine output decreases by up to 30-50%. **Result: overall conservation of water**
Nephrolithiasis due to idiopathic hypercalciuria	Thiazides reduce urinary calcium and may reduce kidney stone recurrence in selected hypercalciuric patients. • Thiazides decrease Ca^{2+} excretion, which lowers Ca^{2+} concentration in the urine, which decreases supersaturation of calcium salts (calcium oxalate and calcium phosphate and crystal formation. Preventive therapy usually consists of a combination of increased fluid intake, dietary modification, weight loss, and drug therapy.

Thiazide Diuretics: Adverse Effects (most are dose-dependent)

ECF volume depletion, mild	Hyperuricemia / gout[3]
Hypotension	Hyperglycemia / Insulin resistance[4]
Hyponatremia[1] (important AE)	Hyperlipidemia[5]
Hypokalemic metabolic alkalosis	GI intolerance (rare)
Hypomagnesemia[2]	Pregnancy: May decrease placental perfusion (caution)
Hypercalcemia (uncommon)	

Mechanisms:

1. NCC blockade impairs urine dilution → impairs free water excretion. Concentrating ability is not impaired so collecting duct responds to ADH release → AQP2 insertion → free water reabsorption + thirst (water consumption) → dilutional hyponatremia. (Recall: TAL / early DCT form the diluting segment.)

   Concomitant NSAID increases risk by enhancing ADH effect (PGs normally counteract ADH.)

2. Chronic thiazide use leads to DCT remodeling and downregulation of apical TRPM6 magnesium channel, which decreases Mg^{2+} reabsorption / increases magnesium wasting.
3. Hypovolemia associated enhanced uric acid reabsorption and reduced secretion via URAT1 (an OAT)
4. Hypokalemia impairs insulin release and thiazides reduce insulin sensitivity → insulin resistance
5. Mild alterations in lipid metabolism related to impaired insulin sensitivity.

DCT: distal convoluted tubule | TAL: thick ascending limb | OAT: organic anion transporter

Thiazide Diuretics: Major Drug Interactions (not a complete list)

Loop diuretics Antihypertensives	**Synergy** – Thiazide diuretics enhance the therapeutic effects. (a favorable drug-drug interaction)
NSAIDs	(1) NSAIDs cause afferent arteriole vasoconstriction → lowers renal blood flow and drug secretion that reduces thiazide delivery to DCT (2) NSAIDs ↑ Na^+/water reabsorption upstream of NCC → **Blunt BP lowering effect of the diuretic** ➢ Remember: *Patients often treat themselves with OTC NSAIDs.*
Lithium	Enhanced Na^+ excretion → compensatory reabsorption of both Na^+ and Li^+ (proximal tubule) → ↑ plasma levels of lithium (narrow therapeutic window) Thiazides can significantly reduce Li^+ clearance → **Lithium toxicity**

Thiazide Diuretics: Major Drug Interactions (continued)

Digitalis glycosides	Hypokalemia →↑ risk of digitalis toxicity (narrow therapeutic window) → **Arrhythmia**
Antidiabetic drugs	Thiazides can blunt the effects of the antidiabetic agent → **Hyperglycemia**
Probenecid	Competition for proximal tubule transporters can reduce thiazide secretion → ↓ **Thiazide efficacy**
Amphotericin B	AMB renal injury → K+ loss due → enhances **potassium-wasting** effect of diuretics
Glucocorticoids	Enhance **potassium-wasting** effect of diuretics

Check your knowledge

How do thiazide diuretics reduce blood pressure for the management of hypertension?

What is the mechanism for the use of thiazide diuretics in the treatment of vasopressin resistance (AVP-R also known as nephrogenic diabetes insipidus), including lithium-induced AVP-R?

What is the mechanism for the thiazide diuretics' calcium preserving effects?

Check your knowledge

What is the mechanism of potassium-wasting and metabolic alkalosis caused by thiazide diuretics?

What is the mechanism for thiazide-induced hyponatremia?

Check your knowledge

What is the mechanism for thiazide-induced insulin resistance?

How does thiazide diuretic therapy increase the risk of lithium toxicity in patients taking lithium?

How does thiazide diuretic therapy increase the risk of digoxin toxicity?

What is the effect of concomitant probenecid on thiazide diuretic therapy?

Check your knowledge

What are the effects of concomitant NSAID and thiazide diuretic therapy on electrolytes and blood pressure?

What are the clinical and mechanistic rationales for using a thiazide diuretic with a loop diuretic?

A 58-year-old female newly diagnosed with primary hypertension is started on hydrochlorothiazide. At follow-up four weeks later, her blood pressure has improved. She reports that she has been trying to follow dietary advice but admits to consuming salted popcorn every afternoon. Routine laboratory testing reveals serum potassium 3.1 mEq/L (normal range 3.5-5.2 mEq/L), which was previously normal. She denies muscle weakness and palpitations.

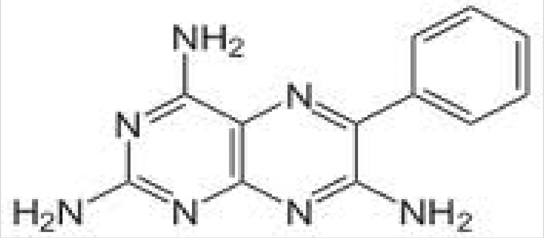
What is the cause of the decreased potassium?

Potassium-sparing diuretics

RENAL EPITHELIUM Na^+ CHANNEL (ENaC) INHIBITORS

- ✳ Amiloride
- Triamterene

Amiloride



Triamterene

ENaC Inhibitors: Pharmacokinetics

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PK Takeaway: ENaC inhibitors are orally absorbed, moderately protein-bound, minimally metabolized (amiloride) or hepatically metabolized (triamterene), renally excreted, and have half-lives long enough to allow once- or twice-daily dosing

Drug	Oral bioavailability Distribution	Onset of diuresis Duration of action	Route of elimination	t½ elimination (approximate)
Amiloride	30-90% Distribution to nephron by glomerular filtration and OCT secretion	2 hours ~24 hours* 1x daily dosing	Urine unchanged drug Feces (unabsorbed fraction)	6-9 hours
Triamterene	30-70% Distribution to nephron by OCT secretion	2-4 hours ~7-9 hours 2x daily dosing	Hepatic metabolism Active metabolite Renal excretion of drug and active metabolite	Not listed t½ of the active metabolite is considerably longer

*Amiloride's long duration of action is related to high potency and slow dissociation from ENaC.

Ion Transport in the Principal Cells

ENaC inhibition

blunts Na^+ reabsorption

↓

↓ intracellular Na^+ concentration

and Na+/K+-ATPase function

↓

the lumen-negative voltage

gradient is neutralized

↓

decreases electrical gradient

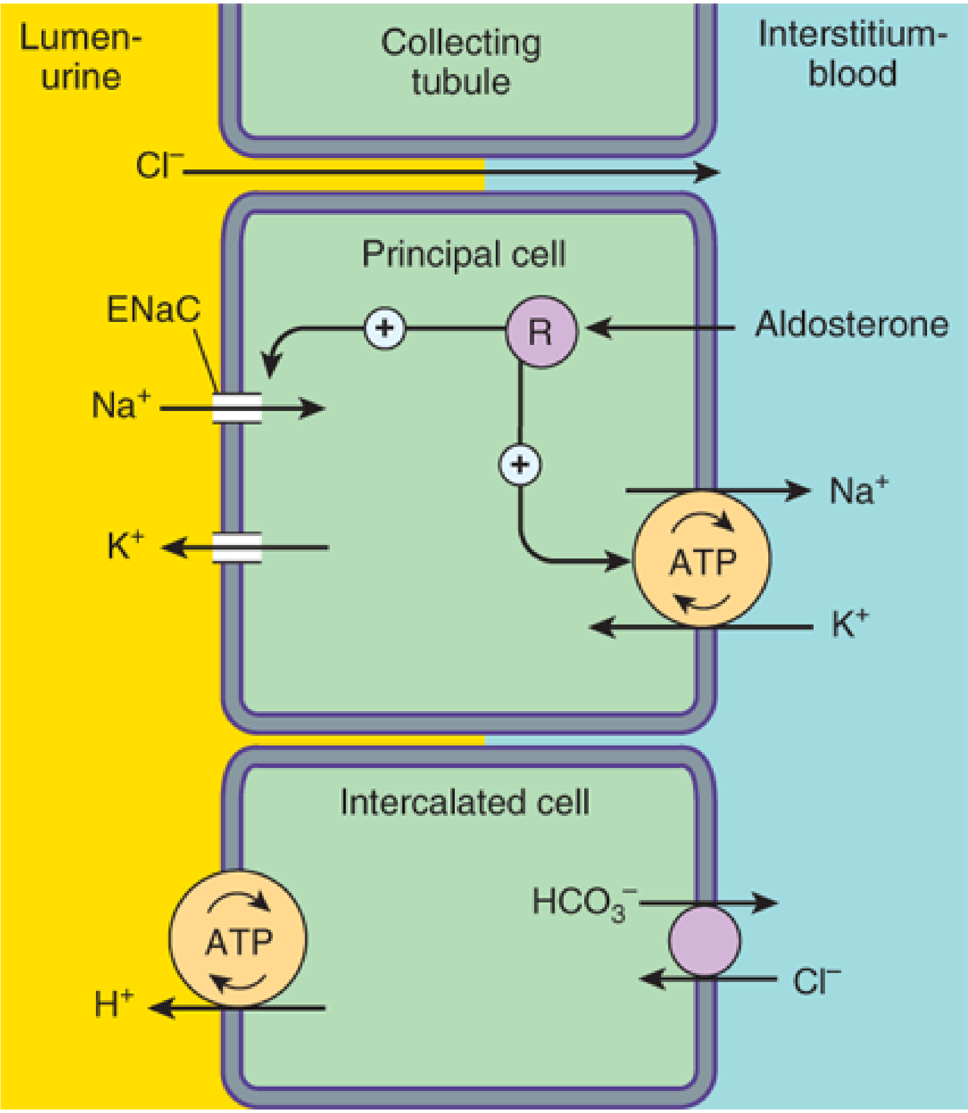
for K+ secretion

↓

↓ K+ secretion

(↑ K+ retention)

K+-sparing effect



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

Physiology

- ~2-5% of filtered Na^+ is reabsorbed in principal cells

Therefore, ENaC block has mild diuretic efficacy.

- ENaC expression is regulated by aldosterone.

Aldosterone increases the activity of **apical membrane channels and Na+/K+-ATPase**.

As Na+ is reabsorbed, the lumen becomes more negative → K+ passively moves into lumen via ROMK channels. This action increases the transepithelial electrical potential with a dramatic increase in both Na+ reabsorption and K+ secretion.

Ion transport pathways across the luminal and basolateral membranes of collecting tubule and collecting duct cells. Inward diffusion of Na^+ via the epithelial sodium channel (ENaC) leaves a lumen-negative potential, which drives reabsorption of Cl^- and efflux of K^+. R, aldosterone receptor.

Citation: Chapter 15 Diuretic Agents, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021 Figure 15-5. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250596112 Accessed: July 23, 2021

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Loop diuretics	++++	0	+++	↑
Thiazides	++	+	++	↑
Loop diuretics plus thiazides	+++++	+	++++	↑
K^+-sparing agents	+	slight	(retained)	↓ mild

+, increase; –, decrease; 0, no change; ↓, acidosis; ↑, alkalosis

ENaC Inhibitors Clinical Uses

Hypertension Edema	ENaC inhibitor + thiazide diuretic combination therapy to: Enhance diuresis Counteract hypokalemia induced by thiazide diuretics
To reduce lithium-induced vasopressin resistance in patients who need to continue lithium therapy	1. Amiloride prevents ENaC-mediated transport of Li+ into principal cells → reduces Li+ accumulation and AQP2 downregulation 2. ENaC + thiazide → Additive effect: Thiazide → mild ECF contraction that increases proximal Na+/water reabsorption → decreases distal Na+/water delivery → lowers urine volume→ reduces polyuria. ***Carefully monitor lithium levels.***

Pathophysiology of lithium-induced vasopressin resistance (nephrogenic diabetes insipidus): Lithium enters principal cells via ENaC and is not transported out. Intracellular accumulation due to chronic lithium exposure leads to principal cell apoptosis, downregulation of AQP2 expression, and collecting duct remodeling. Lithium-induced AVP-R may be irreversible if tubular injury is severe.

Liddle syndrome (pseudoaldosteronism)	ENaC inhibitor + sodium restriction for treatment of hypertension and hypokalemia caused by the disorder

Liddle syndrome is a rare autosomal dominant disorder resulting from a gain of function mutation of genes that regulate ENaC. It affects the C-terminal portion of the β or γ subunit of the ENaC, resulting in unregulated reabsorption of Na^+ and secretion of K^+ and H^+. Plasma renin and aldosterone are suppressed due to Na^+ and fluid retention.

ENaC Inhibitors: Cautions

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Patient factors that increase the risk of hyPERkalemia

- Older age
- High-dose ENaC inhibitor therapy
- Renal impairment
- Hypoaldosteronism
- Combination with other drugs that impair renal K+ excretion

Drug Interactions that increase the risk of hyPERkalemia

- ACEs, ARBs, DRIs

RAS inhibitors ↓ aldosterone secretion which causes ↑ K+ reabsorption

- **NSAIDs:** suppression of PG-dependent renin release leads to ↓ ANG II-mediated aldosterone secretion → ↑ K+ reabsorption
- K+ supplements
- Calcineurin inhibitors (additive hyperkalemic effects)

K+-sparing diuretics are ***contraindicated*** in patients with hyperkalemia or at increased risk of developing hyperkalemia.

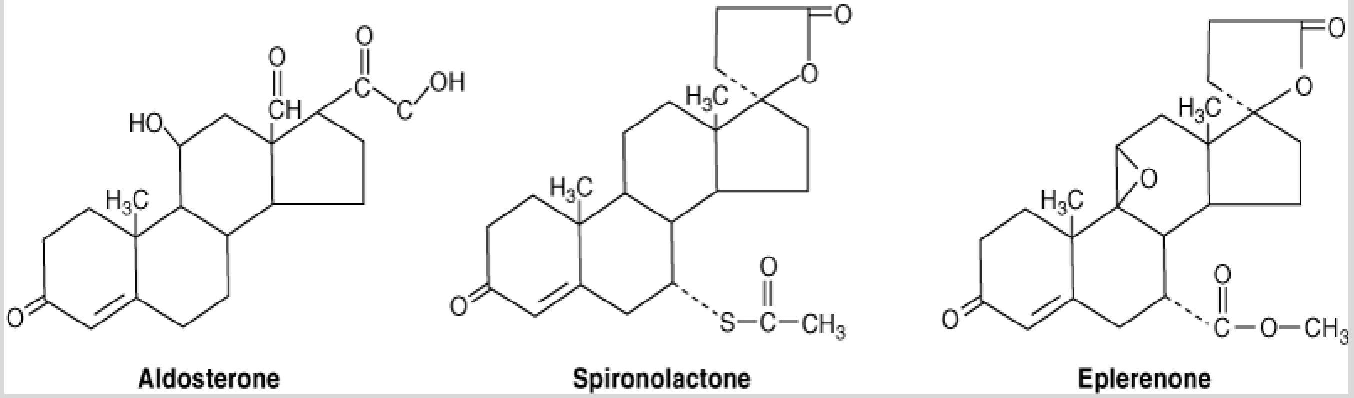
ACE: angiotensin converting enzyme inhibitor
ARB: angiotensin receptor blocker
DRI: direct renin inhibitor
RAS: renin-angiotensin system

Potassium-sparing diuretics

MINERALOCORTICOID RECEPTOR ANTAGONISTS (MRA)

(Aldosterone Antagonists)

- ✳ Spironolactone
- ✳ Eplerenone
- Finerenone
  (nonsteroidal MRA approved for treatment of chronic kidney disease)



Physiology:

Aldosterone is a volume-conserving hormone synthesized in the adrenal cortex from cholesterol.

Aldosterone helps restore blood pressure and circulatory stability in response to hypotension and hypovolemia by acting on high-affinity MRs in the cytosol of principal cells in the collecting ducts.

The MR-aldosterone complex translocates to the nucleus and stimulates gene transcription, which increases Na+-Cl– reabsorption and stimulates K+, H+ secretion.

Pharmacokinetic Properties of MR Antagonists

PK Takeaway: Well absorbed, highly protein bound, hepatic metabolism, renal and biliary excretion, half-lives support once or twice daily dosing.

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	Spironolactone (tablet)	Eplerenone
A	Oral	Oral
D	Protein binding >90%	Protein binding 50% (α1-acid glycoprotein)
M	Hepatic metabolism, rapid, extensive Active metabolites canrenone (most potent) and 7α-spironolactone Enterohepatic circulation **Takeaway:** The active metabolites increase spironolactone's duration of action.	Hepatic CYP3A4 metabolism inactive metabolites
F	~90% with high fat meal	70%
E	Urine / bile, primarily as metabolites	Urine / feces, metabolites
t½	Spironolactone 1-2 h Canrenone 10-23 h 7α-spironolactone 7-20	3-6 hours
T_{max}	3-4 hours (active metabolite)	~1.5 hours
duration	2-3 days (enterohepatic circulation contributes)	7-9 hours

MR Antagonists are competitive antagonists of aldosterone.

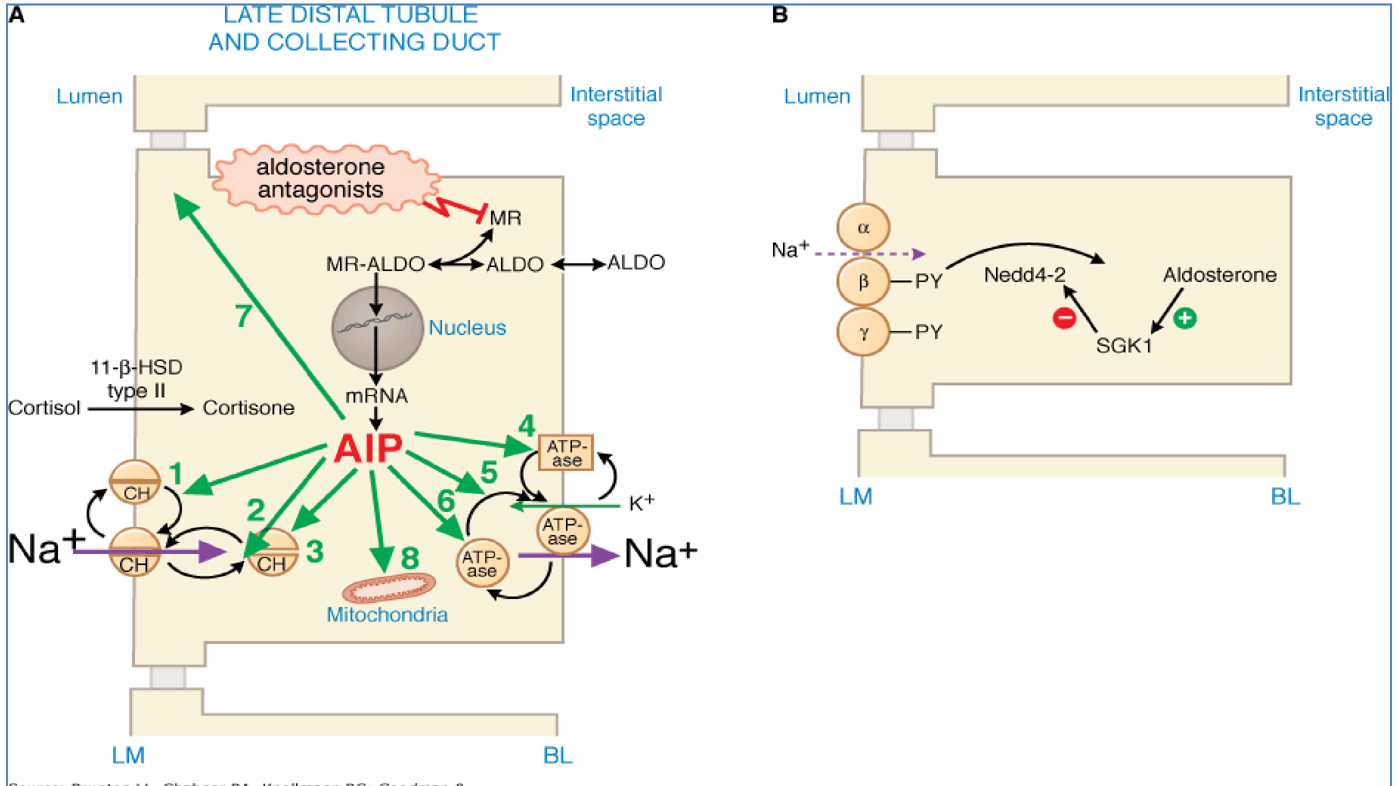
MR antagonists block aldosterone's biological effects on the synthesis and activation of ENaCs, Na+/K+-ATPases and other functions, leading to excretion of Na+ and water, and retention of K+ and H+.

The higher the patient's aldosterone level, the greater the effects of MR antagonists on urinary excretion of electrolytes.

MR-aldosterone complex stimulates transcription of specific genes leading to upregulation of ENaC, N+/K+-ATPase, and ROMK.

Cortisol activates MRs.

Cortisol is inactivated by 11-β-HSD type II, which allows MR activation by aldosterone only.



Left: Effects of aldosterone on late distal tubule and collecting duct and diuretic mechanism of aldosterone antagonists.
AIP: Aldosterone-induced proteins
Right: Nedd4-2 is a ubiquitin ligase that targets ENaC for endocytosis and degradation.
SGK, activated by aldosterone, phosphorylates Nedd4-2 and inactivates it.

Citation: Chapter 25 Drugs Affecting Renal Excretory Function, Brunton LL, Hilal-Dandan R, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e;* 2017 Figure 25-5. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2189&sectionid=170270388 Accessed: July 23, 2021

Spironolactone and Eplerenone: Clinical Uses

Edema Hypertension	MR antagonist is usually combined with a thiazide or loop diuretic for its potassium-sparing effect and added diuretic effect.
Primary aldosteronism: Idiopathic adrenal hyperplasia (bilateral)	Goals of therapy: To raise serum K+ to high-normal range, normalize blood pressure, reverse effects of hyperaldosteronism on heart and kidneys
Heart failure HFrEF and HFpEF patients receiving optimal therapy and low risk for hyperkalemia	MR overexpression in nonepithelial tissues, such as blood vessels and heart, contribute to inflammation, myocardial remodeling and fibrosis, in addition to sodium retention and potassium loss at the distal tubules.
Liver cirrhosis Splanchnic vasodilation causes low arterial volume, which activates RAAS, promoting Na+/water retention, worsening ascites.	Spironolactone (often given with furosemide) Diuretic of choice – long-acting and does not cause hypokalemia Hypokalemia increases ammonia levels by increasing proximal tubule uptake of glutamine (forms ammonia) and upregulation of enzymes involved in ammonia production.
Acne, hirsutism in females Transgender women	Spironolactone: Antiandrogenic effects

Spironolactone and Eplerenone: Cautions

Spironolactone Eplerenone	Hyperkalemia	Other drugs that cause K+ retention: RAS inhibitors, NSAIDs, Calcineurin inhibitors
Spironolactone	Gynecomastia, impotence, loss of libido Androgen receptor antagonist and interference with testosterone biosynthesis shift estradiol/testosterone balance toward estradiol	
Eplerenone is more selective for MRs. Finerenone is MR selective. **Takeaway:** Eplerenone and finerenone do not cause gynecomastia, impotence, or loss of libido.		
Drug interactions	Drugs that increase serum potassium levels.	
Spironolactone	Lab assays: Interferes with assays for detecting serum digoxin levels → false results	
Eplerenone	CYP3A4-mediated interactions Grapefruit juice increases eplerenone AUC ~25% by CYP3A4 inhibition. Manufacturer recommendation: Avoid strong CYP3A4 inhibitors and inducers with eplerenone (and finerenone) therapy.	

Check your knowledge

Based on their mechanisms of action, why are ENaC inhibitors and MR antagonists referred to as weak diuretics?

How do the mechanisms of action of the ENaC inhibitors and MR antagonists differ?

What is the mechanism for the potassium-sparing effect of ENaC inhibitors and MR antagonists?

Check your knowledge

What is the role of potassium-sparing diuretics in the management of hypertension?

What other drugs increase the risk of hyperkalemia when administered with an ENaC inhibitor or an MR antagonist?

Why is an MR antagonist the recommended drug of choice for treating ascites due to hepatic cirrhosis?

Check your knowledge

What is the role of ENaC inhibitors in the treatment of lithium-induced AVP-R (nephrogenic diabetes insipidus)?

Why is an ENaC inhibitor a recommended treatment in Liddle syndrome?

Why is an MR antagonist recommended in HFrEF and HFpEF in patients stabilized on standard therapy?

What are the mechanistic and clinical rationales for treating acne in females with spironolactone?

A 70-year-old woman with hypertension is treated with hydrochlorothiazide and spironolactone. After beginning daily naproxen for knee pain, her labs show rising creatinine and potassium level 5.8 mEq/L (dangerously high).

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

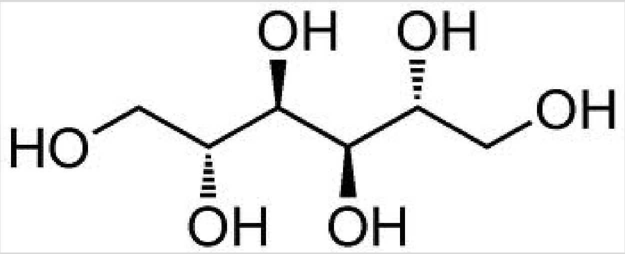
OSMOTIC DIURETICS

✳Mannitol (I.V.)

Urea (I.V.)

Glycerin (Oral)

Isosorbide (Oral)



mannitol

FYI: SGLT2 inhibitors for type 2 diabetes, chronic kidney disease, and heart failure increase urinary excretion of glucose, which acts as an osmotic diuretic and causes modest reduction in plasma volume and blood pressure.

Properties of Osmotic Diuretics

Osmotic diuretics are:

1. Pharmacologically inert
2. Distributed in extracellular fluid
3. Freely filtered in the glomerulus
4. Not reabsorbed by the nephron

Mannitol (I.V.):

- $t_{1/2\,terminal}$ ~5 hours
- Dosed to achieve onset of diuresis in 1-3 hours
- Onset of intracranial pressure (ICP) reduction ~15-30 minutes
- Duration of reduction of ICP 1.5-6 hours

Mannitol is not absorbed from the GI tract.

Osmotic Diuretics Increase Water Excretion

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Site of action: Nephron segments freely permeable to water

- proximal tubule
- descending limb of the loop of Henle

And impairs water reabsorption in the collecting tubule

↑ Urine volume: Osmotically promotes water diuresis

↑ Urinary excretion of:
- Na+
- Nearly all electrolytes

Expand ECF: Distributes in ECF → extracts water from cells

Osmotic Diuretics: Therapeutic Uses

To reduce acute increase in **intracranial pressure with edema** or **intraocular** pressure:

- increases osmotic pressure of plasma → water shift out of brain parenchyma / eye tissue (the water is excreted in urine)
- Caution: Avoid mannitol accumulation in the brain by using intermittent boluses.

As adjunct in some cisplatin regimens to induce forced diuresis.

- Traditionally co-administered to minimize nephrotoxicity
- Note: There is a lack of clear evidence of benefit for this use. Aggressive hydration with IV saline is the preferred approach.

Osmotic Diuretics: Adverse Effects / Cautions

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ECF volume expansion:

Common side effects: Headache, dizziness, nausea, vomiting (common)

Heart failure complication: Florid pulmonary edema as fluid overwhelms the heart's ability to pump it out, especially patients with left-sided heart failure, leading to backup into the lungs, pressure on the pulmonary capillaries, and leakage into alveolar spaces

Hyponatremia: In patients with severe renal impairment, mannitol cannot be excreted → retained in the blood → osmotic extraction of water from cells into vasculature → hyponatremia

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Dehydration and electrolyte imbalance:

Rapid diuresis → loss of water in excess of electrolytes → dehydration and hypovolemia → hypernatremia and hyperkalemia

Profound diuresis (excessive dosing) may cause fluid and electrolyte loss.

Nephrotoxicity: Acute kidney injury (risk factors: pre-existing kidney dysfunction, dose and rate of administration)

Incidence estimated at 6%-7% of patients who receive the drug

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Monitor. Correct electrolyte disturbances. Avoid dehydration.
Discontinue is cardiac or pulmonary status worsens or CNS toxicity develops.

Osmotic Diuretics: Contraindications to Use

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- Allergy to drug or component of formulation
- Severe dehydration
- Severe renal disease (anuria)
- Renal dysfunction after mannitol administration
- Active cranial bleeding (except during craniotomy)
- Progressive heart failure
- Severe pulmonary edema / congestion

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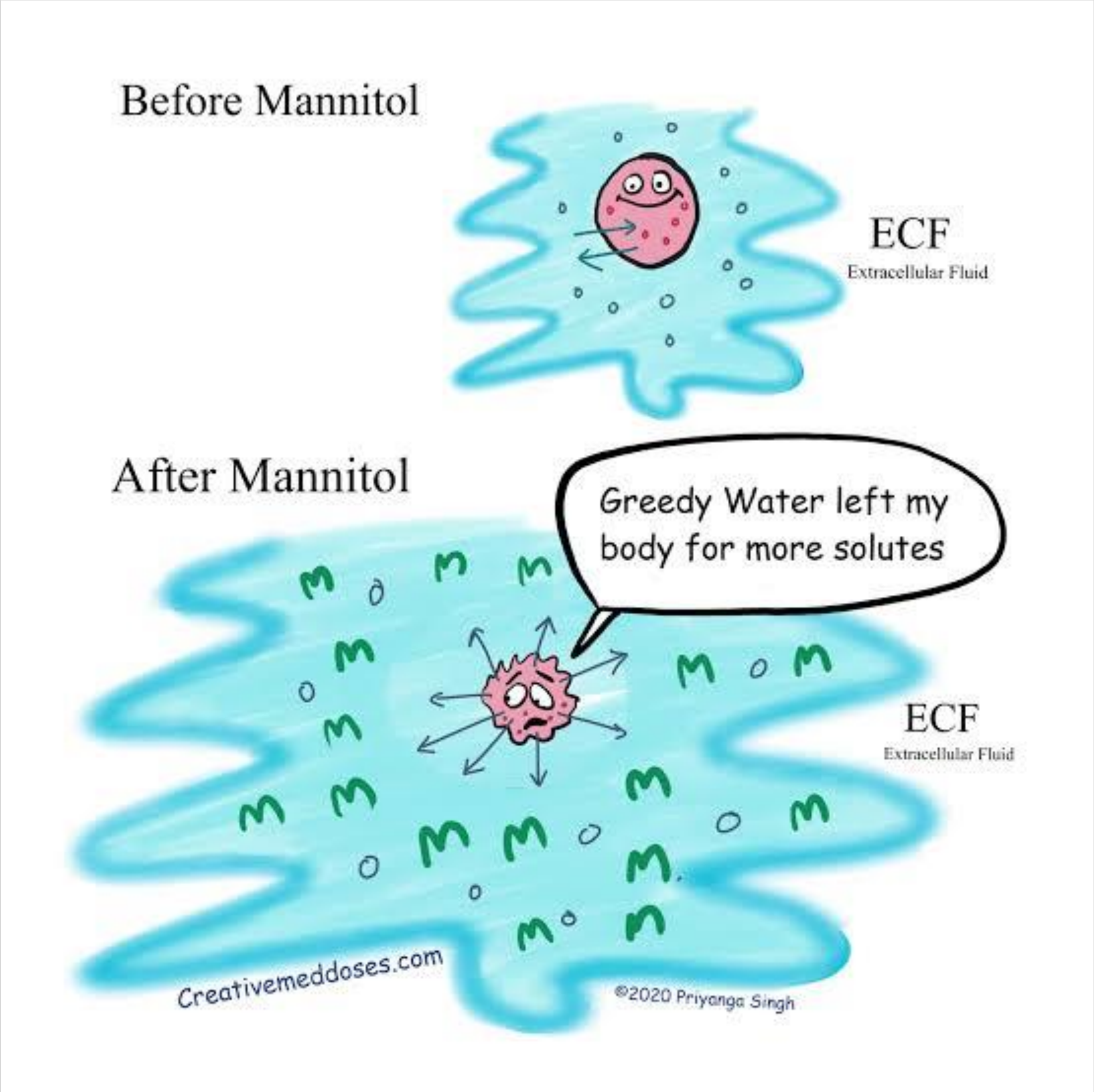
Check your knowledge

What is the primary mechanism of action of mannitol?

Which parts of the nephron is primarily affected by mannitol?

What is mannitol’s primary therapeutic use?



Check your knowledge

What is a potential serious adverse effect of mannitol?

Why is mannitol contraindicated in patients with anuria?

What are the pharmacokinetic properties of mannitol used as a diuretic agent?

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A 65-year-old male is admitted for treatment of a traumatic brain injury and elevated intracranial pressure. He is treated with intravenous mannitol, resulting in initial improvement in neurologic status. Several hours later, he develops acute shortness of breath, crackles on lung exam, and decreased oxygen saturation. Chest imaging reveals bilateral pulmonary edema.

What is the mechanism of this complication?

What risk factors for this complication?

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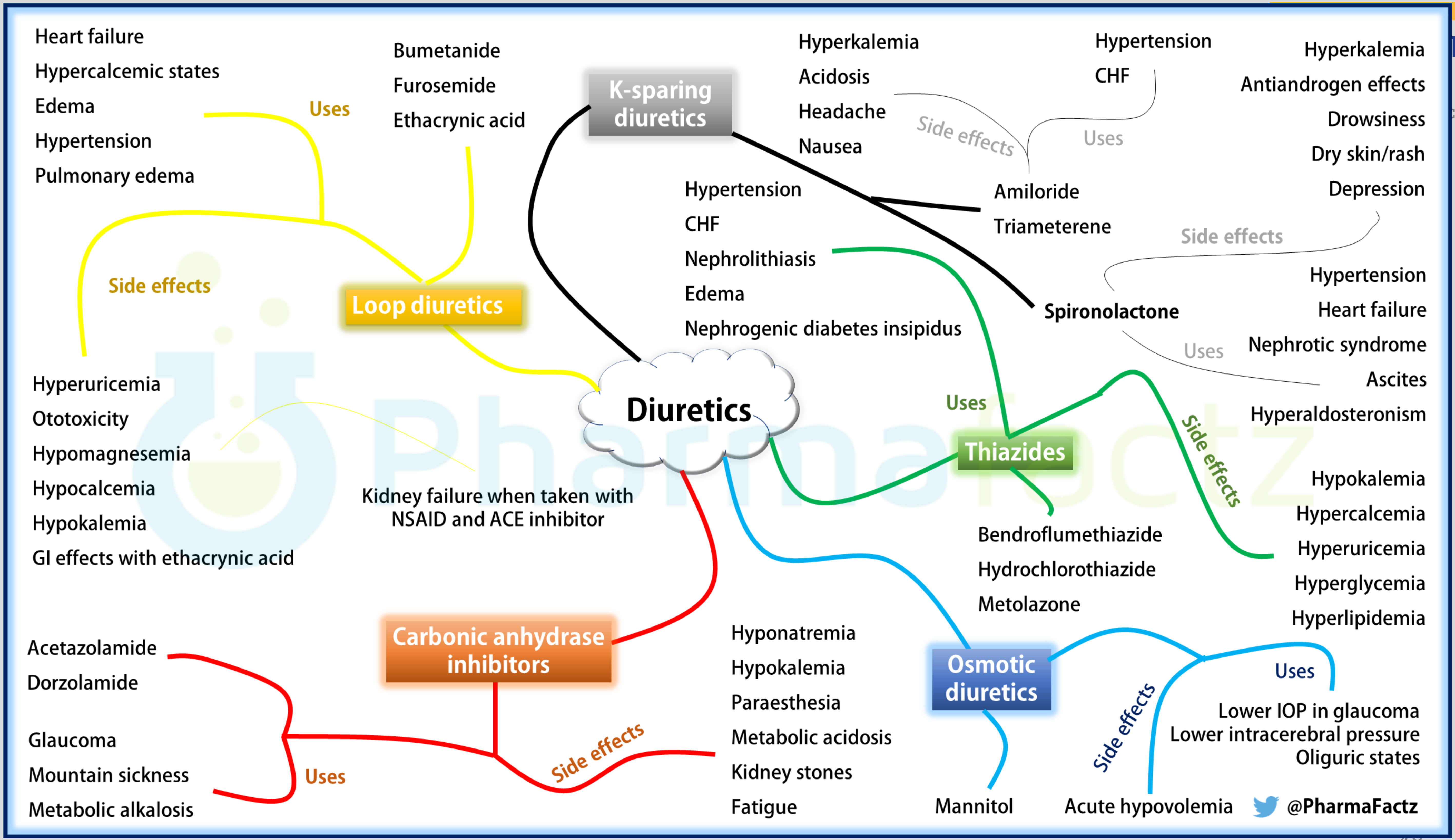
Summary: All diuretics and NSAID-diuretics interactions



https://www.pinterest.com/pin/546131892295889927/

Diuretics sites of action in the nephron segments

- CAIs (PT): weak diuresis, increase bicarbonate loss
- Loop diuetics (NKCC2, TAL): most potent, rapid volume removal, disrupt medullary gradient; impair both concentrating and diluting ability, risk of hyponatremia is lower than with thiazides because dilution is impaired
- Thiazide diuretics (NCC, DCT): moderate natriuresis, impaired diluting ability but not concentrating ability, highest risk of hyponatremia
- K^+-sparing agents (ENaC/aldosterone antagonists, CD): weak natriuresis, prevent K^+ and H+ loss.
- Osmotics (PT/loop of Henle): increase tubular osmotic load throughout the nephron

Major electrolyte and acid-base patterns

- CAIs: ↓ bicarbonate, ↓ Na+, ↓ K^+, ↓ Ca^{2+}, hyperchloremic metabolic acidosis.
- Loops: ↓ Na+, ↓ Cl–, ↓ K^+, ↓ Mg^{2+}, ↓ Ca^{2+}, metabolic alkalosis, mild hyponatremia possible.
- Thiazides: ↓ Na+, ↓ Cl–, ↓ K^+, ↓ Mg^{2+}, ↑ Ca^{2+}, metabolic alkalosis, major cause of diuretic-induced hyponatremia.
- K^+-sparing: ↑ K^+ (risk of hyperkalemia), ↑ H+ / mild metabolic acidosis.
- Osmotics: hypernatremia or hyponatremia depending on water balance.

Volume contraction and compensatory responses to diuretic therapy

- Volume depletion activates RAAS and reduces GFR.
- RAAS activation ↑ aldosterone and ↑ distal Na^+ reabsorption.
- ADH (vasopressin) release causes AQP2 insertion in collecting duct and free water reabsorption.
- Distal nephron compensation limits natriuresis and drives K^+ and H^+ loss.
- *Monitor for hypotension, prerenal azotemia, and renal dysfunction.*

Therapeutic uses

- Edema/CHF: loop diuretics reduce ECF; diuretic resistance potential; loop + thiazide synergy by sequential nephron block.
- Hypertension: thiazides first-line; add ENaC inhibitor or MRA to moderate potassium-wasting effect and increase natriuretic/diuretic effect
- Hypercalcemia: loop diuretics as adjunct after IV hydration – more effective treatments are preferred
- Calcium stones with hypercalciuria: thiazides (selected patients), lifestyle and dietary modifications, weight loss.
- Metabolic alkalosis, mountain sickness, glaucoma: acetazolamide.
- Cirrhosis/ascites: spironolactone first-line – management of hyperaldosteronism.

Drug interactions and clinical cautions

- Thiazides ↑ lithium levels (↑ proximal Na^+/Li^+ reabsorption).
- Loops have a lower risk of lithium toxicity but still require monitoring.
- ACEi/ARBs + K^+-sparing → hyperkalemia risk.
- NSAIDs increase risk of acute kidney injury, worsen edema and heart failure symptoms, increase risk of hyponatremia (thiazides), weaken BP control, blunt diuretic efficacy, worsen metabolic alkalosis.

NSAIDs-Diuretics Interactions and Effects

NSAIDs worsen renal hemodynamics, reduce diuretic delivery, promote sodium retention, and impair potassium excretion.

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Diuretic Class	Mechanism	Resulting Adverse Effect	Why it happens
Thiazides	Block PG-mediated afferent vasodilation	↓GFR Increase AKI risk	Thiazide-induced volume depletion → kidney becomes PG-dependent
	Decrease free water clearance	Hyponatremia	NSAID removes ADH inhibition + thiazide impairs urine dilution
	Decrease diuretic delivery	Effect on BP, edema is blunted	Afferent constriction → ↓RBF reduces GFR → less drug is delivered
Loop diuretics	Afferent constriction → ↓RBF	↓natriuresis diuretic resistance	Afferent constriction → ↓RBF reduces GFR → less drug is delivered
	↑RAAS, ↓PGs	Worsen metabolic alkalosis	Loss of PG-mediate afferent dilation → perceived as low perfusion → RAAS activation →↑K+,H+ secretion, bicarbonate reabsorption
	Sodium retention	Worsen edema, heart failure	NSAIDs promote Na+ retention in TAL/CD
K-sparing diuretics	↓GFR → ↓K+ excretion	Hyperkalemia	NSAIDs and drug ↓K+ excretion – additive
ACEIs / ARBs (RAS inhibitors)	Afferent constriction Efferent dilation	AKI triple hit	NSAID + ACEI/ARB + diuretic → collapse of glomerular filtration pressure
	↓aldosterone, ↓GFR	Severe hyperkalemia	NSAID + RAS inhibitor both ↓K+ secretion

Subclass	**Site of Action**
Carbonic anhydrase (CA) inhibitors *Acetazolamide*	Proximal tubule (primary site) Collecting duct (secondary site)
Loop diuretics NKCC2 ($Na^+/K^+/2Cl^-$ cotransporter) inhibitors *Furosemide, Ethacrynic acid*	Thick ascending limb of the loop of Henle
Thiazide diuretics NCC (Na^+-Cl^- cotransporter) inhibitors *Hydrochlorothiazide, Chlorthalidone*	Distal convoluted tubule
Epithelial Na+ channel inhibitors ENaC inhibitors; K^+-sparing *Amiloride*	Cortical collecting duct principal cells
Mineralocorticoid receptor (MR) antagonists K^+-sparing *Spironolactone, Eplerenone*	Cortical collecting duct principal cells
Osmotic diuretics *Mannitol*	Modifiers of salt and water excretion

Check your knowledge

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How do thiazide diuretics reduce blood pressure for the management of hypertension?

- Acutely lower BP by reducing ECF and cardiac output.
- Sustained effect is due to chronic sodium depletion that decreases vascular smooth muscle responsiveness to vasoconstrictors → sustained reduction in peripheral vascular resistance.

What is the mechanism for the use of thiazide diuretics in the treatment of of vasopressin resistance (AVP-R also known as nephrogenic diabetes insipidus), including lithium-induced AVP-R?

- Thiazide-induced mild volume contraction → RAAS activation and ↓GFR→ Na^+ and water reabsorption in proximal tubule → substantial drop in filtrate reaching ADH-sensitive sites in collecting ducts → overall urine output decreases by up to 30-50% % → new lower steady state is established. Urine output remains chronically lower than baseline (as long as the drug is taken).
- Treatment of AVP-R, including lithium-induced AVP-R: Discontinue lithium (if the patient is taking it) + thiazide diuretic + low sodium, low protein diet + NSAID.

What is the mechanism for the thiazide diuretics' calcium preserving effects?

- Blocking luminal NCC in the DCT lowers intracellular Na^+ concentration→ enhances basolateral NCX1 activity ($3Na^+$ in, 1 Ca^{2+} out), which creates stronger gradient for Ca^{2+} entry via TRPV5 → ↑Ca^{2+} into the cell → exported out via basolateral NCX1. PMCA1b (ATPase) fine-tunes Ca^{2+} export (a secondary mechanism).

Check your knowledge

What is the mechanism of the potassium-wasting and hypokalemic metabolic alkalosis caused by thiazide diuretics?

- Blocking NCC in the DCT increases sodium and water delivery to the collecting ducts.
- Delivery of an increased sodium load to the aldosterone-sensitive sites in the distal nephron along with volume contraction → aldosterone (due to RAAS activation) upregulates ENaC and Na^+/K^+ ATPase → increases sodium reabsorption by ENaC → enhances the lumen negative potential → increases K+ excretion → potassium-wasting effect
- K+ loss / hypokalemia, increased H+ secretion, ECF volume contraction / RAAS activation → hypokalemic metabolic alkalosis. Hypochloremia and bicarbonate retention sustain metabolic alkalosis.

What is the mechanism for thiazide-induced hyponatremia?

Thiazide diuretics impair the kidney's ability to dilute urine, while concentrating ability remains intact. **Retention of water is greater than retention of sodium → dilutional hyponatremia.**

- (1) **NCC inhibition impairs urine dilution:** Free water is generated in the diluting segments by removing solute without water. NCC inhibition decreases NaCl reabsorption, thereby increasing solute concentration, which impairs excretion of free water.
- (2) **Urine concentrating ability is preserved:** The medullary gradient remains intact, and the collecting duct responds to ADH → AQP2 insertion in collecting ducts → free water reabsorption.
- Increased thirst tends to increase fluid intake.

Most patients do not develop hyponatremia unless water excretion is impaired.

Check your knowledge

What is the mechanism for thiazide-induced insulin resistance?

- Hypokalemia disrupts the function of the ATP-sensitive K+ channels in pancreatic beta cells, leading to impaired insulin release (primary mechanism) + thiazides reduce tissue utilization of glucose (impaired insulin sensitivity) → hyperglycemia.

How does thiazide diuretic therapy increase the risk of lithium toxicity in patients taking lithium?

- Blocking NCC increases Na+ excretion → compensatory reabsorption of both Na+ and Li+ in the proximal tubule → ↑ plasma levels of lithium (narrow therapeutic window) → even a modest increase in serum lithium levels can lead to lithium toxicity.

How does thiazide diuretic therapy increase the risk of digoxin toxicity?

- Hypokalemia augments digitalis-mediated increase in cytosolic calcium levels → intracellular Ca^{2+} overload → increased cardiac automaticity → arrhythmia
- FYI: Digoxin competes with K+ binding to the Na+/K+ ATPase in cardiomyocytes. When K+ is low (hypokalemia), more digoxin binds → stronger inhibition → digoxin toxicity

What is the effect of concomitant probenecid on thiazide diuretic therapy?

- Probenecid is a potent inhibitor of organic anion transporters (OATs). Diuretics, including thiazide diuretics enter the nephron predominantly via OATs. Blocking the OATs reduces the concentration of diuretic in the nephron, which can reduce diuretic efficacy.

Check your knowledge

What are the effects of concomitant NSAID and thiazide diuretic therapy on electrolytes and blood pressure?

NSAIDs reduce the blood pressure lowering effect of thiazide diuretics by:

- (1) Afferent arteriole vasoconstriction that lowers renal blood flow and drug secretion, which reduces thiazide delivery to DCT
- (2) ↑ Na^+/water reabsorption upstream of NCC

What are the clinical and mechanistic rationales for using a thiazide diuretic with a loop diuretic?

Reduction of edema by sequential nephron blockade for synergistic natriuretic/diuretic effect.

- Loop diuretics block sodium reabsorption in the NKCC2 in the TAL. Some of the sodium is reabsorbed at distal sites in the nephron.
- Thiazide diuretics block NCC in the DCT, which is distal to the TAL, therefore preventing reabsorption of the sodium load in the DCT.

A 58-year-old female newly diagnosed with primary hypertension is started on hydrochlorothiazide. At follow-up four weeks later, her blood pressure has improved. She reports that she has been trying to follow dietary advice but admits to consuming salted popcorn every afternoon. Routine laboratory testing reveals serum potassium 3.1 mEq/L (normal range 3.5-5.2 mEq/L), which was previously normal. She denies muscle weakness and palpitations.

What is the cause of the decreased potassium?

- Thiazides increase distal sodium delivery.
- Higher sodium intake increases distal sodium delivery, leading to increased ENaC-mediated sodium reabsorption and potassium secretion, resulting in hypokalemia.

Check your knowledge

Based on their mechanisms of action, why are ENaC inhibitors and MR antagonists referred to as weak diuretics?

- Approximately 2-5% of sodium excretion occurs in the principal cells. Therefore, ENaC inhibitors and MR antagonists block only a small fraction of the total sodium, thus water, reabsorption by the kidneys.

How do the mechanisms of action of the ENaC inhibitors and MR antagonists differ?

- ENaC inhibitors block sodium reabsorption through the luminal epithelial sodium channels.
- MR antagonists pass through the basolateral membrane and block the mineralocorticoid receptor in the cytosol. The complex translocates to the nucleus, binds specific response elements on the DNA, which stimulates transcription and expression of various genes involved in the sodium reabsorption and potassium excretion.
- Blocking either site of action reduces Na+ reabsorption / increases Na+ excretion, followed by water.

What is the mechanism for the potassium-sparing effect of ENaC inhibitors and MR antagonists?

- Blocking Na+ reabsorption in the principal cells abolishes the transepithelial potential difference, which reduces the electrochemical gradient for potassium. K+ is retained, which can lead to hyperkalemia.

Check your knowledge

What is the role of potassium-sparing diuretics in the management of hypertension?

- ENaC inhibitors and MR antagonists are frequently combined with a thiazide diuretic to counteract the potassium-wasting effect of the thiazide diuretic and for added diuretic effect.

What other drugs increase the risk of hyperkalemia when administered with an ENaC inhibitor or an MR antagonist?

- RAS inhibitors (ACE inhibitors, ARBS, and direct renin inhibitors), NSAIDs, potassium supplements, and calcineurin inhibitors

Why is an MR antagonist the recommended drug of choice for treating ascites due to hepatic cirrhosis?

- Spironolactone is drug of choice in the treatment because of its aldosterone antagonism effects.

Mechanism:

- Portal hypertension increases hydrostatic pressure, which leads to fluid transudation into the peritoneal cavity, which contributes to effective hypovolemia.
- Splanchnic vasodilation is triggered by nitric oxide and other vasodilators in response to portal hypertension, which leads to reduced systemic vascular resistance, which activates SNS and RAAS and vasopressin leading to sodium and water retention.
- Hypoalbuminemia due to liver dysfunction decreases oncotic pressure, which facilitates fluid leakage from the capillaries into the interstitial and peritoneal spaces.

Check your knowledge

What is the role of ENaC inhibitors in the treatment of lithium-induced AVP-R (nephrogenic diabetes insipidus)?

- ENaC inhibition counteracts vasopressin resistance by preventing Li+ entry to principal cells.
- Lithium enters principal cells via ENaC and is not pumped out efficiently. LI+ accumulation in the principal cells interferes with expression and trafficking of AQP2. Thus, the collecting duct becomes impermeable to water. ENaC inhibition prevents Li+ entry to principal cells.

Why is an ENaC inhibitor a recommended treatment in Liddle syndrome?

- Liddle syndrome is a rare autosomal dominant disorder resulting from a gain of function mutation of genes that regulate ENaC.
- ENaC inhibitor + sodium restriction for treatment of hypertension and hypokalemia caused by the disorder

Why is an MR antagonist recommended in HFrEF and HFpEF in patients stabilized on standard therapy?

- Evidence supports the use of spironolactone in conjunction with standard therapy in patients at low risk for hyperkalemia by blocking the effects of aldosterone overexpression.

What are the mechanistic and clinical rationales for treating acne in females with spironolactone?

- Spironolactone has antiandrogen effects and shifts estrogen/testosterone balance toward estrogen.

A 70-year-old woman with hypertension is treated with hydrochlorothiazide and spironolactone. After beginning daily naproxen for knee pain, her labs show rising creatinine and potassium level 5.8 mEq/L (dangerously high).

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Triple hit:

- Thiazides reduce effective circulating volume → ↓ renal perfusion pressure. Therefore, compensatory mechanisms maintain GFR, including prostaglandin-mediate afferent dilation.
- NSAIDs block renal prostaglandin synthesis, leading to afferent arteriolar constriction, which reduces renal blood flow, therefore reduces GFR.
- Thiazide + NSAID together cause a ***marked reduction in GFR***. Decreased GFR increases serum creatinine and risk of acute kidney injury.
- Spironolactone decreases ENaC activity and Na+ reabsorption, which removes the lumen-negative potential, therefore decrease K+ secretion via ROMK. Potassium is retained.
- NSAIDs-induced inhibition or renin release decreases aldosterone-mediated secretion of K+ → K+ retention

Mechanism	Effect
Thiazide	Volume depletion
NSAID	↓afferent dilation → ↓GFR
Spironolactone	↓K+ secretion
↓ GFR	↓K+ excretion

Check your knowledge

What is the primary mechanism of action of mannitol?

- Mannitol (a sugar alcohol) and other osmotic diuretics increase the concentration of solutes in the plasma and interstitial spaces and in the renal tubules.
- The elevated osmolarity produces an osmotic gradient, which draws water from areas of lower solute concentration, like the brain parenchyma and eye, into the plasma and interstitial fluid and the renal tubules. The water is excreted as urine.

Which parts of the nephron is primarily affected by mannitol?

- The proximal tubule and descending limb of the loop of Henle are permeable to water. Mannitol osmotically draws water into the nephron at these segments. Mannitol also impairs water reabsorption in the collecting tubule.

What is mannitol's primary therapeutic use?

- Intracranial pressure (ICP). By increasing solute concentration in plasma, mannitol osmotically draws water out of the brain parenchyma into the intravascular space, reducing ICP. The water is excreted in the urine.
- Intraocular pressure (IOP): By the same mechanism, mannitol reduces IOP in acute emergencies where rapid IOP reduction is critical such as acute angle-closure glaucoma and optic trauma.

Check your knowledge

What is a potential serious adverse effect of mannitol?

- Volume expansion leading to pulmonary edema in patients with left-side heart failure as fluid overload causes the fluid backs up into the lungs, increasing pulmonary capillary pressure, causing leakage into alveolar spaces.
- Volume expansion causing electrolyte imbalance with hyponatremia due to fluid overload or hypernatremia.
- Dehydration with rapid water loss greater than electrolyte loss leading to hypernatremia and hyperkalemia.
- Profound diuresis causing fluid and electrolyte loss.

Why is mannitol contraindicated in patients with anuria?

- In patients not making urine (anuria), mannitol accumulates in the extracellular fluid, leading to volume overload and pulmonary edema, potentially hypertensive crisis, hypertensive encephalopathy, renal failure.

What are the pharmacokinetic properties of mannitol used as a diuretic agent?

- Mannitol is administered intravenously; dosed to achieve rapidly distributes to extracellular compartment; freely filtered by the glomerulus; minimal reabsorption (stays in the nephron); onset of ICP reduction in about 15 to 30 minutes; duration of ICP reduction 1.5-6 hours.

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What is the mechanism of this complication?

- Mannitol stays in the extracellular space, which increases plasma osmolality.
- The osmotic gradient pulls water from the intracellular to the interstitial to the intravascular compartment, which causes rapid expansion of plasma volume.
- Increased intravascular volume increases venous return to the heart (increases preload), which increases pulmonary capillary pressure.
- High hydrostatic pressure in the pulmonary capillaries pushes fluid into the lung tissue, causing pulmonary edema.

What risk factors for this complication?

- Patients with heart failure, renal impairment, or existing volume overload cannot handle sudden volume expansion.

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Resources

- Access Medicine Goodman & Gilman The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 29: Regulation of Renal Function and Vascular Volume
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